Ashish Shajan

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Robotics & Automation Engineer with strong hands-on experience in AI-driven systems, robotic manipulation, and autonomous navigation. Specialized in ROS 2, SLAM, PX4, MAVROS, reinforcement learning, and sensor fusion. Experienced in simulation tools such as CARLA, NVIDIA Isaac Sim, Gazebo, and IPG CarMaker. Skilled in delivering autonomous robotics solutions bridging simulation and real-world deployment.

PROFESSIONAL EXPERIENCE

Chief Technical Officer — AT Bots **Jan 2025 – Present**

Engineer (Jan – Apr 2025), Chief Technical Officer (Apr 2025 – Present) Kerala, India

- Architected a ROS 2-based autonomous drone stack integrating PX4 and MAVROS for real-time flight control and navigation.
- Designed and implemented multi-sensor fusion (IMU, GNSS, LiDAR) pipelines for accurate localization and obstacle avoidance.
- Directed system-level validation, simulation testing (Gazebo/PyBullet), and hardware deployment for AI-powered structural inspection solutions.
- Represented AT Bots at the FutureX 2025 Finals at Google (Top 10 startups in India, Google Cloud × Campus Fund).

Captain, Autonomous Head & Finance Head — Team Equinox, SAE aBAJA **Sep 2024 – Aug 2026**

Autonomous Systems Engineer (Sep 2024) → Autonomous Head (Mar 2025) → Captain, Autonomous Head & Finance Head (2026 season) Kottayam, India

- Led the end-to-end development of a Level 2 autonomous BAJA vehicle built from the ground up, driving technical roadmap, milestone planning, and competition strategy.
- Coordinated a 30+ member cross-functional team (mechanical, electrical, embedded, AI) to ensure seamless hardware–software integration and track readiness.
- Architected a modular ROS 2-based autonomous stack integrating perception, sensor fusion (Radar, IMU, wheel encoders), localization (EKF), and control modules (ACC, AEB, LKA).
- As Finance Head (2026), owned the team budget, purchase approvals, sponsorship and vendor payments.
- Oversaw simulation workflows in Gazebo, CARLA, and IPG CarMaker prior to real-world deployment, and real-time integration and performance tuning on track.

Research Intern — Robotics Research Centre, IIIT Hyderabad **Oct 2024 – Aug 2025**

Hyderabad, India

- Developed high-fidelity Gazebo simulation environments and a PX4 + ROS 2 offboard-control and GPS-failsafe test workspace to validate autonomous drone control strategies prior to hardware testing.
- Engineered a custom autonomous mobile robot from scratch, implementing 2D LiDAR-based SLAM to generate high-resolution occupancy grid maps.
- Developed real-time differential drive kinematics and wheel odometry estimation pipelines; performed HITL testing and PID tuning.

Intern — Intel Unnati Industrial Training Program, Intel Corporation **May – Jul 2024**

Bengaluru, India (Hybrid)

- Selected for Intel's national-level industrial training program focused on advanced AI and data-driven system development.
- Designed data preprocessing, feature engineering and model validation pipelines; built a heart-attack risk prediction model (86.8% accuracy, F1 0.88).
- Presented technical findings and solution strategies through structured reports and demonstrations.

PROJECTS

12-DOF Quadruped Robot (ROS 2, Isaac Sim, Gazebo) **2024 – 2025**

- Designed and built a 12-DOF quadruped with a ROS 2 locomotion framework: forward/inverse kinematics, crawl and trot gaits with ros2_control, and IMU-based posture correction.
- Validated in NVIDIA Isaac Sim and Gazebo (joint torques, stability margins), then deployed on a 3D-printed, servo-driven robot; tuned PID joint controllers.

BAJA SAE India: Autonomous Software Stack (ROS 2, Control Systems, Computer Vision) **2025 – 2026**

- Radar-based perception and EKF fusion (IMU, wheel encoders); ADAS functions ACC, AEB and LKA in ROS 2; CAN-based drive-by-wire interface nodes; validated in Gazebo, CARLA and IPG CarMaker.

Crack Detection with Drones using AI (PyTorch, ROS 2, PX4) **2025**

- Deep learning crack detection optimized for edge devices, integrated with autonomous drone navigation (PX4 + MAVROS + ROS 2) for automated structural inspection.

Differential Drive Mobile Robot (ROS 2, Raspberry Pi, RPLiDAR) **2025**

- 2D LiDAR SLAM with Cartographer, real-time differential drive kinematics and wheel odometry, validated in structured indoor environments.

Mini projects: Waste segregation with ML (Carbon Bytes), manipulator and self-balancing robot with roboRIO, underwater surveillance system.

SKILLS

Robotics & Navigation: Autonomous Vehicles, Control Systems, Inverse Kinematics, Autonomous Mobile Robots, Differential Drive Kinematics, Nav2, SLAM (LiDAR), Localization, Odometry, Sensor Fusion (IMU+Encoders)

AI & Machine Learning: Deep Learning, Computer Vision, PyTorch, YOLOv8, U-Net++, EfficientNet, Model Optimization

Frameworks & Middleware: ROS 2, PX4 Autopilot, Gazebo, RViz, micro-ROS

Software: Gazebo, Isaac Sim, IPG CarMaker, RViz, CARLA, PyBullet, MATLAB, QGroundControl

Embedded Systems: ESP32, Raspberry Pi, Teensy, NVIDIA Jetson, PID Control, Trajectory Generation

Programming & Tools: Python, C++, Embedded C, Linux, Git, Docker, PlatformIO

Leadership & Management: Technical Leadership, Project Planning, Budgeting & Finance, Team Collaboration, System-Level Thinking, Decision-Making, Clear Communication

EDUCATION

B.Tech in Robotics and Automation — Saintgits College of Engineering (Autonomous), Kottayam, Kerala **2022 – 2026**

CGPA: 7.6 · Relevant courses: Control Systems, Robot Kinematics & Dynamics, AI, Machine Learning, Embedded Systems, Computer Vision, Mobile Robots, Robot Operating System.

Higher Secondary — Jawahar Navodaya Vidyalaya, Wayanad

CERTIFICATIONS & ACHIEVEMENTS

- Top 10 startups in India — FutureX 2025 Finals at Google (Google Cloud × Campus Fund), as CTO of AT Bots.
- 3rd Place, Vruthi 2025: Clean Kerala Conclave Wasteathon (Rs. 25,000) for AI-powered waste segregation.
- 3rd Place, HackS'US Edition IV national hackathon, Rajagiri School of Engineering & Technology (Rs. 15,000).
- Intel Unnati Industrial Training — 2-month internship program.
- NPTEL Certification: Soft Skill Development.

LANGUAGES

English (Professional) · Malayalam (Native) · Hindi (Intermediate)